

题目

已知 $E^\circ(\text{Pb}^{2+}/\text{Pb}) = -0.1266 \text{ V}$, $E^\circ(\text{PbO}_2/\text{Pb}^{2+}) = 1.458 \text{ V}$, $K_{sp}(\text{PbSO}_4) = 1.8 \times 10^{-8}$, $K_{a2}(\text{H}_2\text{SO}_4) = 1.0 \times 10^{-2}$

请计算铅酸蓄电池的标准电动势，选出最接近的选项。

A. 1.5 V

B. 1.6 V

C. 1.7 V

D. 1.8 V

E. 1.9 V

F. 2.0 V

G. 2.1 V

H. 2.2 V

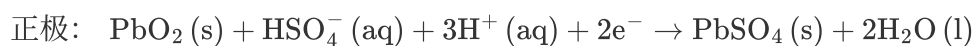
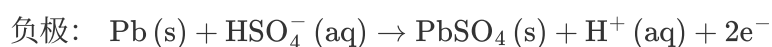
答案

正确答案: **E**

详细解析

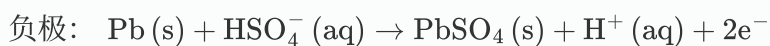
这是一道物理化学题目，考察电化学中能斯特方程的计算。

先写出正负极半反应：



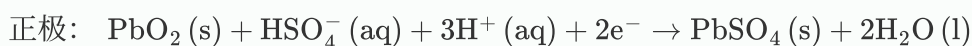
CHECKPOINT

0.5 PTS



CHECKPOINT

0.5 PTS



首先对负极进行讨论。已知 $E^\circ(\text{Pb}^{2+}/\text{Pb}) = -0.1266 \text{ V}$ ，对应半反应方程式 $\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- = \text{Pb(s)}$ ，则 $E^\circ(\text{PbSO}_4/\text{Pb}) = E^\circ(\text{Pb}^{2+}/\text{Pb}) + \frac{RT}{2F} \ln[\text{Pb}^{2+}]$

下求解 $[\text{Pb}^{2+}]$ ：

$$[\text{Pb}^{2+}] = \frac{K_{sp}(\text{PbSO}_4)}{[\text{SO}_4^{2-}]}$$

$$\text{又} [\text{SO}_4^{2-}] = \frac{K_{a2}[\text{HSO}_4^-]}{[\text{H}^+]} = 1.0 \times 10^{-2} \text{ mol/L}$$

CHECKPOINT

1 PTS

$$[\text{SO}_4^{2-}] = 1.0 \times 10^{-2} \text{ mol/L}$$

$$\text{代入} [\text{Pb}^{2+}] = \frac{K_{sp}(\text{PbSO}_4)}{[\text{SO}_4^{2-}]} = \frac{1.8 \times 10^{-8}}{1.0 \times 10^{-2}} = 1.8 \times 10^{-6} \text{ mol/L}$$

CHECKPOINT

1 PTS

$$[\text{Pb}^{2+}] = 1.8 \times 10^{-6} \text{ mol/L}$$

$$\text{则 } E^\circ(\text{PbSO}_4/\text{Pb}) = E^\circ(\text{Pb}^{2+}/\text{Pb}) + \frac{RT}{nF} \ln[\text{Pb}^{2+}] = -0.296 \text{ V} \quad (\text{代入 } T = 298\text{K}, \text{ 后同})$$

CHECKPOINT

1 PTS

$$E^\circ(\text{PbSO}_4/\text{Pb}) = -0.296 \text{ V}$$

下面对正极进行讨论。已知 $E^\circ(\text{PbO}_2/\text{Pb}^{2+}) = 1.458 \text{ V}$ ，对应半反应方程式 $\text{PbO}_2(\text{s}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- = \text{Pb}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$ ，则

$$E^\circ(\text{PbO}_2/\text{PbSO}_4) = E^\circ(\text{PbO}_2/\text{Pb}^{2+}) + \frac{RT}{2F} \ln \frac{[\text{H}^+]^4}{[\text{Pb}^{2+}]}$$

$$\text{同理, } [\text{Pb}^{2+}] = 1.8 \times 10^{-6} \text{ mol/L}$$

$$\text{代入, } E^\circ(\text{PbO}_2/\text{PbSO}_4) = E^\circ(\text{PbO}_2/\text{Pb}^{2+}) + \frac{RT}{2F} \ln \frac{[\text{H}^+]^4}{[\text{Pb}^{2+}]} = 1.62 \text{ V}$$

CHECKPOINT

1 PTS

$$E^\circ(\text{PbO}_2/\text{PbSO}_4) = 1.62 \text{ V}$$

因此，电动势为 $1.62 - (-0.296) = 1.92 \text{ (V)}$

CHECKPOINT

0.5 PTS

电动势与 1.9 V 更接近